MIL-C-22211A(EC)
5 September 1969
SUPERSEDING
MIL-C-22211(SHIPS)
30 October 1959

MILITARY SPECIFICATION

CALIBRATOR SET, RADIO, AN/UDM-1( )

1. SCOPE

1.1 This is a general performance specification which covers one type of test equipment for calibration of radiac survey equipment.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-601-Boxes, Wood, Cleated-Plywood.

MILITARY
MIL-P-116-Preservation, Methods of.
MIL-P-15024-Plates, Identification——Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-E-15090-Enamel, Equipment, Light-Gray(Formula No. 111).
MIL-B-15395-Brazing Alloys, Silver.
MIL-E-16400-Electronic Equipment, Naval Ship and Shore: General Specification.
MIL-M-19590-Marking of Commodities and Containers to Indicate Radioactive Material.
MIL-I-45208-Inspection System Requirements.

STANDARDS

MILITARY
MIL-STD-129-Marking for Shipment and Storage.
MIL-STD-1186-Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; with Appropriate Test Methods.

DRAWINGS

BUREAU OF SHIPS
REF 10085-Shipping Crate for Source Container.
REF 60001-AN/UDM-1( ) Radiac Calibrator Set, Modified.
REF 60001-2-AN/UDM-1( ) Modified Source Container for.
REF 60001-3-Base Assembly.
REF 60001-4-Stand Assembly.
RED 60001-5-Plate Sub-Assembly.
RED 60001-6-Tripod Assembly.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Blvd., Chicago, Ill. 60606.)

NATIONAL CLASSIFICATION BOARD
National Motor Freight Classification Rules

(Application for copies should be addressed to the National Motor Freight Traffic Association, Inc., 1616 P Street, N.W., Washington D.C. 20036.)

DEPARTMENT OF TRANSPORTATION
Code of Federal Regulations, Title 49
Department of Transportation Specification DOT-55

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.)

3. REQUIREMENTS

3.1 First article sample.-Prior to beginning production a sample shall be inspected as specified in 4.3 (see 6.2).

3.2 General description.-The Radiac Calibrator Set AN/UDM-1( ) is a calibrating set which provides a suitable radioactive source for calibrating radiac survey equipment.

3.2.1 The Radiac Calibrator Set AN/UDM-1( ) shall be constructed and assembled in accordance with this specification and Drawings REF 60001 and REF 60001-2. Where the requirements of this specification conflict with the drawings, this specification shall govern. Units are to be constructed to comply with with Department of Transportation (DOT) Specification 55 and all Radiac Transportation Permits.

3.3 Material.-Materials specified herein and in Drawings REF 60001 and REF 60001-2 shall be entirely suitable for the purpose intended. Use of other material shall have the approval of the command or agency concerned.

3.3.1 Silver solder shall conform to grade III of MIL-B-15395.

3.3.2 Lead shall be at least 95.0 percent pure.

3.4 Equipment composition. - The calibration equipment shall consist of the following:

(a) Radiation source - Cesium 137.
(b) Source container.
(c) Platform for source container.
(d) Rolling platform for radiac equipment.
(e) Track or rails for the rolling platform.
(f) Optical system.

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3.4.1 Radiation source. - The source holder shall be designed and fabricated to meet Department of Transportation "Special Form" requirements and be compatible with the AN/UDM-1( ) Source Container. The source activity shall be 120 curies of Cesium 137 ±5% with no more than 3% Cesium 134 permissible. The true activity, when the source is installed in the AN/UDM-1( ) shall be at least 40 R/hr at one meter.

3.4.1.1 Procurement requirements for sealed sources. - Source holder must be designed and tested so as to allow U. S. (AEC) Atomic Energy Commission licensing for possession and use of the source and this includes the following information and tests, which shall be supplied to the Naval Electronic Systems Command, Code 05162 16 months after date of contract. In the event of first article sample waiver, the information required shall be delivered 3 months after date of contract.

(a) The source model number.

(b) A drawing or sketch of the source holder describing all materials of construction, dimensions, and methods of sealing the source.

(c) A facsimile of the label to be affixed to the source or source holder and a description of how and where the label will be permanently attached.

(d) A description of tests performed on prototype sources that establish the integrity of the source construction and seal under the most adverse conditions of use to which the source is likely to be subjected.

(e) A description of the quality control procedures to be followed to assure that each finished source meets specifications furnished to the Naval Electronic Systems Command. A description of tests for contamination and leakage of radioactive material should be included.

If tests and other information have already been indexed or furnished to U. S. AEC only source drawing and model number are necessary. The contractor shall certify that the source meets DOT requirements for special form.

3.4.1.2 Permit requirements. - In the event a permit is required, the contractor shall make application for permits through the cognizant Military Traffic Management and Terminal Service Office via the DCAS Transportation office administering contract. Permits when required will be obtained prior to the movement of shipment. Permit number (when applicable) will be shown on DD-250's and Shipping Documents. The date of issue, date of expiration and office issued from will also be shown.

3.4.2 Source container. - The arrangement and construction of the source container shall be as shown on Drawing REF60001-2.

3.4.2.1 Radiographic requirement. - Prior to shipment, each source container shall be radiographed, and certified by the contractor to be free of holes or other defects in the lead.

3.4.3 Platform for source container. - The overall size of the platform and the size and shape of the individual parts shall conform to Drawing REF 60001-3.

3.4.3.1 The components and materials used in the fabrication shall be such that an applied force of 1000 pounds will not cause buckling or overstressing of individual parts or the complete assembly.

3.4.4 Rolling platform for radiac equipment. - Construction of the rolling platform shall conform to Drawing REF 60001-4. This platform is not only a supporting structure for radiac equipment but also for parts of the optical system (RED 60001-5). The following requirements, in addition to the requirements as given in Drawings REF 60001-4 and RED 60001-5, shall govern the design and construction.

3.4.4.1 The size and shape of the components (major units) and the material used for fabrication shall be such that the structure is light yet strong enough so that the application of a 50-pound load will not cause buckling, deflections or overstressing of the individual parts or the structure as a whole.

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3.4.4.2 The vertical axis of the rolling platform shall remain perpendicular at all points on the track.

3.4.4.3 The rollers and bearings shall have such characteristics that they will carry the load easily and allow the operator to move the platform along the track with a minimum of effort.

3.4.4.4 Means shall be provided to keep the rolling platform on the track at all times.

3.4.4.5 A clamping or holding device shall be incorporated which permits the rolling platform to be locked to the track at any point.

3.4.4.6 The design of the jig and radiac equipment supporting platform shall be such that it can be raised or lowered easily through the distance specified. A lock shall hold the platform securely and firmly in any position.

3.4.4.7 The rolling platform shall be constructed in such a manner that it can be readily disassembled into several subassemblies and thus allow the whole unit to be packed into a minimum of space.

3.4.5 Track. - The track or rails shall be built in 5-foot sections. The individual sections shall be designed and constructed for easy assembly and remain in alignment with respect to each other. Construction design shall be in accordance with Drawing REF 60001-3.

3.4.5.1 The track length shall be 30 feet.

3.4.5.2 The entire track shall have a straight, smooth top surface.

3.4.5.3 There shall be no buckling or deflecting of the rails at any point under load.

3.4.6 Optical system (tripod assembly). - Construction of the tripod assembly shall be in accordance with Drawing RED 60001-6.

3.5 External finish. - External finish of the equipment shall be as specified in MIL-E-16400. The finish coats shall be grey enamel in accordance with type II, class 2, of MIL-E-15090.

3.6 Markings. - A calibration plate shall be attached to the source container having the following inscription:

Calibrated by_____
Date of calibration _____
Unit No._____

3.6.1 The unit number and calibrated by, and date of calibration of the source will be inserted by the National Bureau of Standards, upon calibration of the source.

3.6.2 In addition, an identification plate in accordance with type A or B of MIL-P-15024 shall be attached to the source container.

3.7 Workmanship. - Workmanship shall be as specified in MIL-E-16400.

3.8 Safety. - The design, development and production shall promote maximum safety of both operational and maintenance personnel and equipments during all phases of operational life in accordance with MIL-E-16400.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except

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as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor's quality assurance system. - The contractor shall provide and maintain an effective inspection and quality assurance system, acceptable to the Government, covering the supplies under the contract. The quality assurance system shall be in accordance with the requirements of MIL-I-45208. A current written description of the system shall be submitted to the cognizant Government inspector, for approval, sixty (60) days prior to first article inspection. Any changes to the approved plan which might affect the degree of assurance required by this specification or other applicable documents shall be submitted to the cognizant inspector and approved in writing prior to use.

4.1.2 Government verification. - All quality assurance operations performed by the contractor will be subject to Government verification at any time. Verification will consist of (a) surveillance of the operations to determine that practices, methods, and procedures of the written system description are being properly applied, (b) Government product inspection to measure quality of product to be offered for acceptance, and (c) Government product inspection of delivered products to assure compliance with all requirements of this specification. Failure of the contractor to promptly correct deficiencies discovered by him or of which he is notified shall be cause for suspension of acceptance until corrective action has been made or until conformance of product to prescribed criteria has been demonstrated.

4.1.3 Test procedures. - The contractor shall submit written test procedures including test equipment and tolerance limits to be used for first article inspection and production inspection to the command or agency concerned, via the Government inspector for review and approval sixty (60) days prior to beginning tests. Each test shall be identified in accordance with paragraph 4.2 as to classification.

4.1.3.1 Procedures for inspections, tests, demonstrations, etc. shall contain step by step information which will enable the operator to properly perform the requirements of this specification. The method of examination shall be clearly defined.

4.1.3.2 The procedures shall contain a list of test equipment identified by name and model. Special test equipment shall also be listed and identified.

4.1.3.3 Data sheets shall be included in the procedures. Conformance parameters shall be clearly indicated with tolerance limits. Whenever quantitative measurements are made, the result shall be shown by quantitative data.

4.2 Classification of inspection. - The method of examination and testing of the equipment fall within the following classifications:

(a) First article inspection
(b) Production inspection
(c) Shipping inspection

4.3 First article inspection. - First article inspection shall be made at a Government laboratory designated by the command or agency concerned on the prototype model submitted by the contractor to determine suitability for Naval use and shall consist of all examination and testing necessary to determine compliance with the requirements of this specification. First article inspection shall include the tests specified in table I.

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TABLE I. First article inspection

Requirement	Reference
Surface examination	4.6.1
Operating test	4.6.2
Structural integrity	4.6.3

4.4 Production inspection. - Production inspection shall be made on every equipment offered for delivery. The inspection shall comprise such examination and testing as will prove the workmanship and reveal the omissions and errors of the production process such as functional and performance tests at a limited number of points, tests which detect deviations from design, tests of adjustment, and tests which detect hidden defects of material. Production inspection shall consist of the tests approved in accordance with paragraph 4.1.3 and shall include the requirements of table II.

TABLE II. Production inspection

Requirement	Reference
Surface examination	4.6.1
Operating test	4.6.2

4.5 Shipping inspection. - Inspections shall be conducted to insure conformance with the requirements of Section 5 of this specification.

4.6 Inspection procedures. -

4.6.1 Surface examination. - Equipment shall be examined for the following:

(a) Workmanship, assembly and fit, mechanical safety, and marking.
(b) Materials, parts and finish.

For conformance to 3.2, 3.3, 3.4.1, 3.4.2, 3.4.3, 3.4.4, 3.4.5, 3.4.6, 3.5, 3.6, 3.7 and 3.8.

4.6.2 Operating test. - The equipment, with the source in place, shall be subjected to an operating test to insure qualitatively the proper functioning of the equipment including:

(a) All operating controls and adjustments
(b) True activity (using a Victoreen R meter or equivalent)
(c) Radiographic integrity
(d) Safety requirements

For conformance to 3.4.1, 3.4.2.1, 3.4.4.2 through 3.4.4.7 and 3.8.

4.6.3 Structural integrity. - The equipment shall be tested to determine conformance with 3.4.3.1, 3.4.4.1 and 3.4.5.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging. - Packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. -

5.1.1.1 Source containers. - Each source container shall be individually packaged in accordance with Method II of MIL-P-116. The aperture plug control arm system shall be lubricated for normal operation.

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5.1.1.2 Optical system. - Each optical system shall be individually packaged in accordance with submethod IA-8 of MIL-P-116 without preservative compounds.

5.1.1.3 Other equipment. - Other equipment such as platforms and tracks or rails shall be individually packaged in accordance with method III of MIL-P-116.

5.1.1.4 Technical literature. - Technical Literature shall be packaged in transparent, waterproof, heat sealed, plastic bags, minimum 4 mil thick and shall not be placed within the barrier used to package any other item.

5.1.2 Level C. - Each complete set shall be packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. - Packing shall be level A, B or C as specified (see 6.1).

5.2.1 Level A. -

5.2.1.1 Source container. - Each source container shall be packed in a container conforming to Drawing REF 10085 and Department of Transportation Regulations. Cushioning, anchoring, bracing and blocking shall be in accordance with MIL-STD-1186 to prevent movement and damage during shipment and handling. The top of the source container shall be blocked so that it can not become separated from the source container. The plug control arm shall be blocked so that it can not move during shipment and handling.

5.2.1.2 Accessory equipment. - Accessory equipment comprising a complete set, packaged as specified in 5.1 shall be packed in containers conforming to PPP-B-601, overseas type without cleats on the top panel. Cushioning, anchoring, bracing, blocking and waterproofing shall be in accordance with MIL-STD-1186. The gross weight of shipping containers shall not exceed 200 pounds.

5.2.2 Level B. - Each source container and accessory equipment packaged as specified in 5.1 shall be packed as specified in 5.2.1, except that PPP-B-601 boxes may be domestic type and weight limitations for accessory equipment packs shall be as permitted by PPP-B-601. Domestic type PPP-B-601 boxes shall be strapped as specified in the appendix.

5.2.3 Level C. - Items comprising a complete set, except for the source container which shall be packed in accordance with 5.2.1.1, shall be packed in containers acceptable to the common carrier and which will insure safe delivery at destination in a satisfactory condition at the lowest applicable rate. Containers, packing or method of shipment shall comply with Uniform Freight and National Motor Freight Classification Rules or regulations of other carrier rules as applicable to the mode of transportation.

5.3 Marking. - In addition to any marking required by the contract or order, unit packages, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Radioactive equipment. - Each item, package and shipping container containing radioactive material shall be marked in accordance with MIL-M-19590 and specifically Code of Federal Regulations, Title 49, Section 73.391 through 73.394, as applicable. Each shipping container shall be marked with "fragile" and "arrow" markings.

5.3.1.1 Unpacking instructions. - Radioactive source, shipping containers shall be provided with unpacking instructions as follows:

> "Reusable container for Radioactive Source. Retain for reuse. To open, remove screws with screwdriver and topcover. Remove screws and shipping brackets holding down housing base inside container. Leave housing on base until metal stand is assembled."

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Unpacking instructions shall be placed in a sealed waterproof envelope marked "UNPACKING INSTRUCTIONS" and affixed to the outside of the shipping container in a protected location, preferably between the cleats on the end of the container adjacent to the identification marking.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Levels of preservation, packaging, packing, and marking (see section 5).

6.2 First article. - Invitations for bids should provide that the Government reserves the right to waive the requirement for first article samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

Preparing activity:
Navy - EC

(Project 6665-N254)

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

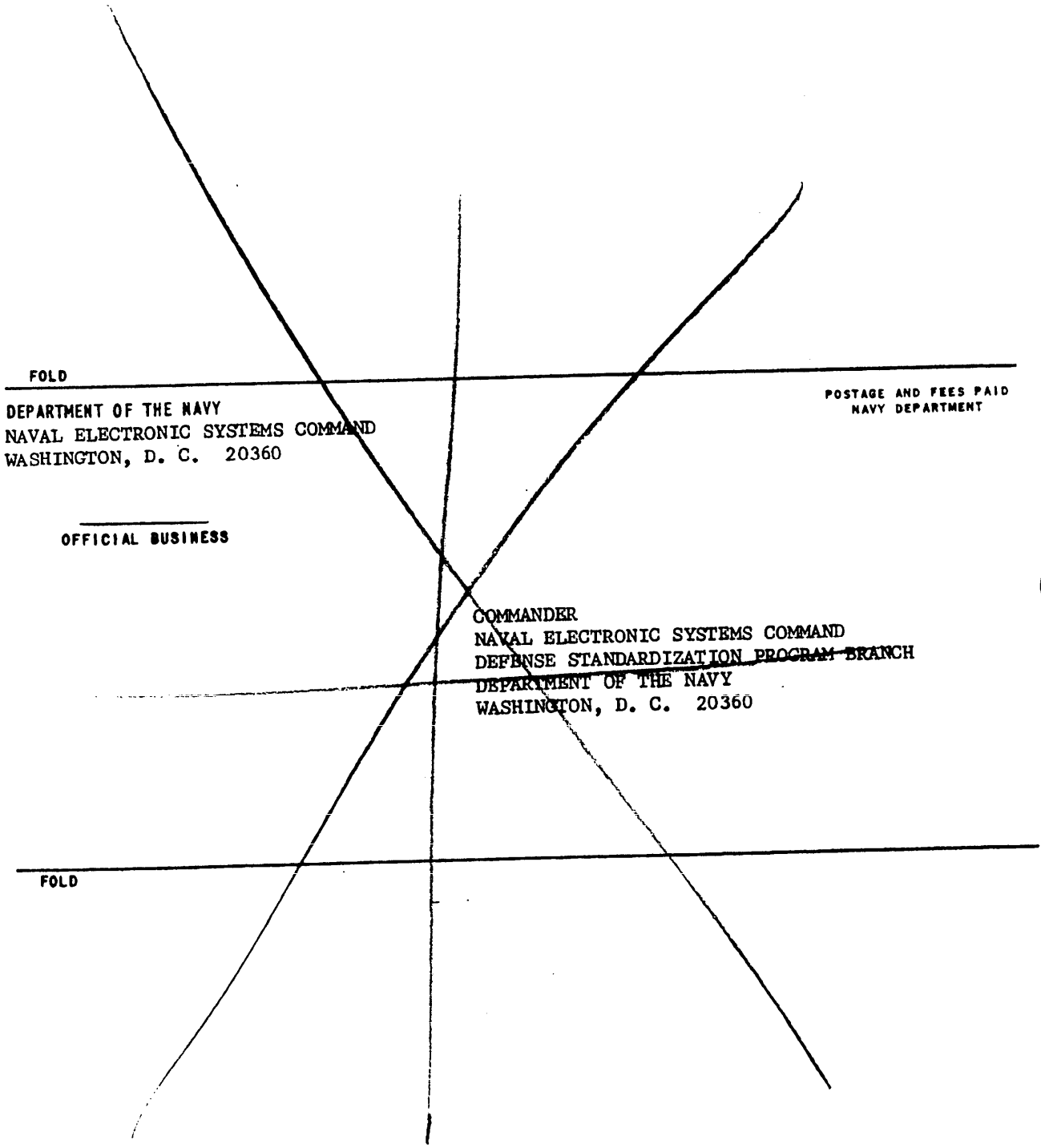
4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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